A smooth-block tiling framework for the upper bound in Erdős Problem 302, with exactly checkable certificates

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Abstract

Let $f(N)$ be the largest size of a set $A \subseteq \{1, \dots, N\}$ containing no three distinct elements with $1/a = 1/b + 1/c$ (Erdős Problem 302). We describe an elementary framework that proves upper bounds for $f(N)/N$ by decomposing $\{1, \dots, N\}$ into disjoint dilates of a fixed finite block of smooth numbers, and by lower-bounding the number of deletions needed in each block with fractional packing certificates in exact rational arithmetic. With a $\{2, 3\}$-block it recovers the constant $25/28 \approx 0.8929$; with a $\{2, 3, 5, 7, 11\}$-block it gives

$$f(N) \le (0.8638\ldots + o(1))N,$$

with an exact rational constant that is checked by a short standard-library Python script. *These results are weaker than the bound* $0.84617\ldots$ *announced in September 2026 by K. Kitamura with a Lean proof, which uses the same smooth-block decomposition at a much larger scale; we do not improve the state of the art.* The contribution of this note is a self-contained exposition of the single-level tiling special case, explicit certificates, and a record of where this simple method saturates. The exact order of $f(N)/N$ remains undetermined; it is known only to lie between $5/8$ and the bounds above (the value $\frac{1}{2}$ is already excluded by Cambie's construction).

1 Introduction

Call a triple $\{a, b, c\}$ of distinct positive integers *harmonic* if $\frac{1}{a} = \frac{1}{b} + \frac{1}{c}$. Erdős Problem 302 [1] asks for the size $f(N)$ of a largest $A \subseteq \{1, \dots, N\}$ containing no harmonic triple, and whether $f(N) = (\frac{1}{2} + o(1))N$. According to the problem page, the sets of all odd integers in $[1, N]$ and of all integers in $(N/2, N]$ give $f(N) \ge (\frac{1}{2} + o(1))N$, S. Cambie improved this to $f(N) \ge (\frac{5}{8} + o(1))N$ (all odd integers up to $N/4$ together with all integers in $[N/2, N]$), and W. van Doorn proved $f(N) \le (\frac{9}{10} + o(1))N$ [1].

On the page for the closely related Problem 301 [2], van Doorn's argument is recorded as follows: the sets $S_a = \{2a, 3a, 4a, 6a, 12a\} \cap [1, N]$, with a ranging over the integers $8^b 9^c d$ with $\gcd(d, 6) = 1$, are pairwise disjoint, and any admissible set must omit at least two elements of S_a when $a \le N/12$ and at least one element when $N/12 < a \le N/6$. The fact that at least two elements of $\{2, 3, 4, 6, 12\}$ must be omitted already follows from the three harmonic triples $\{2, 3, 6\}, \{3, 4, 12\}, \{4, 6, 12\}$ (no single element meets all three), so the same computation

applies to Problem 302 and gives the constant 25/28 (we derived this ourselves from the quoted argument rather than finding it stated for Problem 302); Section 3 recovers it as the simplest instance of our framework.

In September 2026, K. Kitamura posted a Lean 4 proof of $\limsup f(N)/N \le 0.8461739827964010$ [3]. According to the repository's description, the proof also decomposes $[1, N]$ into blocks $q \cdot S_P(N/q)$ of P-smooth numbers, but obtains the lower bounds for the cover number of $S_P(x)$ through a recursion over a chain of 20 prime sets (adding the primes up to 71), with tables on $[0, 500000]$ and more than a thousand branch-and-bound certificates. That repository states that its proof has not yet been reviewed independently by a human expert; we have not re-run it, and we take its statement at face value only in order to place our results.

Proposition 1 (Cambie's construction)**.** *The set* $A = \{n \le N/4 : n \text{ odd}\} \cup \{n : N/2 \le n \le N\}$ *contains no harmonic triple. In particular* $f(N) \ge (\frac{5}{8} + o(1))N$*, so the value* $\frac{1}{2}$ *asked about on the problem page is not the correct asymptotic density.*

Proof. Let $a < b < c$ be harmonic, so $a = \frac{bc}{b+c}$ and $c = \frac{ab}{b-a}$. If $b, c \ge N/2$ then $a = \frac{bc}{b+c} \ge \frac{N}{4}$, with equality only for $b = c = N/2$; since $b \ne c$ we get $a > N/4$, and also $a < \frac{N}{2}$ because $\frac{bc}{b+c} \le \frac{N}{2}$ with equality only for $b = c = N$. So $a \notin A$. Otherwise $b < N/2$, hence b and $a < b$ are odd integers $\le N/4$; then ab is odd and $b - a$ is even, so $c = \frac{ab}{b-a}$ is not an integer. Both cases are impossible. □

What this note does. We isolate the *single-level tiling* version of the smooth-block idea (Theorem 2): a fixed box of exponent vectors tiles the exponent lattice, and a certified lower bound on the deletions needed inside the box, as a function of the truncation level, yields a closed-form density. The result is a bound that is easy to state, easy to check exactly, and weaker than the state of the art. We include certificates and a checker so that every number below can be reproduced without trusting floating-point software.

2 The framework

Fix distinct primes $P = \{p_1, \dots, p_r\}$ and integers $a_1, \dots, a_r \ge 1$. Let

$$\mathcal{C} = \Big\{ p_1^{e_1} \cdots p_r^{e_r} : 0 \le e_i < a_i \Big\}, \qquad U = \Big\{ p_1^{a_1 k_1} \cdots p_r^{a_r k_r} : k_i \ge 0 \Big\}.$$

Every $n \in \mathbb{N}$ has a unique representation $n = d\,u\,c$ with $\gcd(d, p_1 \cdots p_r) = 1$, $u \in U$, $c \in \mathcal{C}$ (the p_i-adic valuation v_i of n is written uniquely as $a_i k_i + e_i$). Hence $\mathbb{N}$ is the disjoint union of the *blocks*

$$\beta(d, u) = \{duc : c \in \mathcal{C}\}, \qquad \gcd(d, P) = 1,\ u \in U.$$

The harmonic equation is invariant under scaling: if $\{a, b, c\}$ is harmonic then so is $\{ma, mb, mc\}$ for every $m \ge 1$.

For $z \ge 1$ let $\mathcal{H}(z)$ be the 3-uniform hypergraph on the vertex set $\mathcal{C}$ whose edges are the harmonic triples $\{a, b, c\} \subseteq \mathcal{C}$ with $\max\{a, b, c\} \le z$, and let $g(z)$ be its covering number (the minimum size of a set of vertices meeting every edge). Clearly g is non-decreasing. We allow any non-decreasing integer-valued step function γ with $\gamma(z) \le g(z)$ for all z and $\gamma(z) = 0$ for $z < 1$, and write

$$\tau_k = \inf\{z : \gamma(z) \ge k\} \qquad (k = 1, \dots, K), \quad K = \sup \gamma,$$

so that $\gamma(z) = \#\{k : \tau_k \le z\}$.

Theorem 2. *For every $N \ge 1$,*

$$f(N) \le N - \sum_{k=1}^{K} \sum_{u \in U} \#\big\{d \ge 1 : \ \gcd(d, P) = 1, \ u\, d\, \tau_k \le N\big\}. \tag{1}$$

Consequently

$$f(N) \le (1 - D)\, N + O_{P,a,\gamma}\big(N^{1/2}\big), \qquad D = \prod_{i=1}^{r} \frac{1 - 1/p_i}{1 - p_i^{-a_i}} \sum_{k=1}^{K} \frac{1}{\tau_k}.$$

Proof. Let $A \subseteq [1, N]$ contain no harmonic triple and put $D_N = [1, N] \setminus A$, so D_N meets every harmonic triple contained in $[1, N]$. Fix a block $\beta(d, u)$ and put $z = N/(ud)$. An element duc of the block lies in $[1, N]$ iff $c \le z$. By scaling invariance, every harmonic triple $\{a, b, c\} \subseteq \mathcal{C}$ with $\max \le z$ gives a harmonic triple $\{dua, dub, duc\} \subseteq [1, N]$ inside the block. Hence $D_N \cap \beta(d, u)$, divided by du, is a cover of $\mathcal{H}(z)$, and $|D_N \cap \beta(d, u)| \ge g(z) \ge \gamma(z)$. The blocks are disjoint, so

$$|D_N| \ge \sum_{d,u} \gamma\Big(\frac{N}{ud}\Big) = \sum_{k=1}^{K} \sum_{u \in U} \#\{d : \gcd(d, P) = 1, \ ud\, \tau_k \le N\}$$

(with τ_k as above; the last equality is $\gamma(z) = \#\{k : \tau_k \le z\}$). Since $f(N) = N - \min |D_N|$, this is (1).

For the asymptotics, write $\varphi = \prod_i (1 - 1/p_i)$. By inclusion–exclusion over subsets of P, $\#\{d \le x : \gcd(d, P) = 1\} = \varphi x + \theta$ with $|\theta| \le 2^r$. Fix k. Summing over $u \in U$ with $u \le N/\tau_k$ gives

$$\sum_{\substack{u \in U \\ u \le N/\tau_k}} \Big(\frac{\varphi N}{u \tau_k} + \theta_u\Big) = \frac{\varphi N}{\tau_k}\Big(S - R_k\Big) + O\big(2^r \#\{u \in U : u \le N\}\big),$$

where $S = \sum_{u \in U} u^{-1} = \prod_i (1 - p_i^{-a_i})^{-1}$ and $R_k = \sum_{u > N/\tau_k} u^{-1}$. By Rankin's trick, $R_k \le (N/\tau_k)^{-1/2} \sum_{u \in U} u^{-1/2} = O(N^{-1/2})$, because $\sum_{u \in U} u^{-1/2} = \prod_i (1 - p_i^{-a_i/2})^{-1} < \infty$. Moreover $\#\{u \in U : u \le N\} = O(\log^r N)$. Summing over k gives $|D_N| \ge \varphi S\, N \sum_k \tau_k^{-1} - O(N^{1/2}) = DN - O(N^{1/2})$. □

Certificates

Lemma 3. *Let $\mathcal{E}$ be the edge set of a hypergraph and let $w : \mathcal{E} \to \mathbb{Q}_{\ge 0}$ satisfy $\sum_{e \ni x} w_e \le 1$ for every vertex x. Then every cover H has $|H| \ge \sum_e w_e$. In particular, if $\sum_e w_e > k - 1$ for an integer k, then every cover has at least k vertices.*

Proof. $\sum_e w_e \le \sum_e w_e |e \cap H| = \sum_{x \in H} \sum_{e \ni x} w_e \le |H|$. □

To use Theorem 2 we therefore proceed as follows. For $k = 1, 2, \ldots$, we find the smallest threshold τ_k such that the sub-hypergraph $\mathcal{H}(\tau_k)$ carries a fractional packing $w^{(k)}$ (Lemma 3) of total weight $> k - 1$. Then $\gamma(z) = \#\{k : \tau_k \le z\} \le g(z)$ and D is an explicit rational number. In the computations below, the packings come from a floating-point LP solver, are rounded *down* to multiples of 10^{-9}, and are re-verified in exact integer arithmetic (loads $\le 10^9$, total $> (k-1)10^9$); only the verified rational data enter the theorem.

3 Results

Theorem 4. *The following hold, with the constants and certificates verified in exact arithmetic by the script* `verify.py` *supplied with this note.*

1. $P = \{2, 3\}$, *box* $(3, 2)$ *(cells* $1, 2, 3, 4, 6, 12$*):* $K = 2$, $(\tau_1, \tau_2) = (6, 12)$, $D = 3/28$, *hence* $f(N) \le (\frac{25}{28} + o(1))N$.

2. $P = \{2, 3, 5\}$, *box* $(6, 5, 3)$ *(90 cells, 385 triples):* $K = 30$, $D = \frac{3315}{26257}$, *hence* $f(N) \le (0.87374\ldots + o(1))N$.

3. $P = \{2, 3, 5, 7, 11\}$, *box* $(7, 5, 3, 2, 2)$ *(420 cells, 5601 triples):* $K = 140$, $D = \frac{6227587}{45732192}$, *hence*
$$f(N) \le (0.863824\ldots + o(1))N.$$

Hand-checkable proof of part (1). The harmonic triples in $\mathcal{C} = \{1, 2, 3, 4, 6, 12\}$ are $\{2, 3, 6\}$, $\{3, 4, 12\}$, $\{4, 6, 12\}$ (indeed $\frac{1}{2} = \frac{1}{3} + \frac{1}{6}$, $\frac{1}{3} = \frac{1}{4} + \frac{1}{12}$, $\frac{1}{4} = \frac{1}{6} + \frac{1}{12}$). The triple $\{2, 3, 6\}$ has maximum 6, the other two have maximum 12. A single triple needs one deleted element, so $\tau_1 = 6$ (packing weight 1 on $\{2, 3, 6\}$). For $z \ge 12$ take weight $\frac{1}{2}$ on each of the three triples: the loads are $\frac{1}{2}$ on 2 and 1 on each of $3, 4, 6, 12$, so the total $\frac{3}{2} > 1$ gives $\gamma \ge 2$, i.e. $\tau_2 = 12$. Here $U = \{8^i 9^j\}$, so $\sum_u u^{-1} = \frac{8}{7} \cdot \frac{9}{8} = \frac{9}{7}$ and $\varphi = \frac{1}{3}$; thus $D = \frac{1}{3} \cdot \frac{9}{7} \cdot (\frac{1}{6} + \frac{1}{12}) = \frac{3}{28}$. □

Proof of parts (2) and (3). Finite exact verification of the certificates $w^{(k)}$ (Lemma 3), followed by Theorem 2; see `verify.py`. □

Sanity checks against exact values. For $N \le 68$ the block bound of Theorem 4 never exceeds the number of deletions $N - a(N)$ implied by the first 68 terms of OEIS A390395 [4]. For larger N we computed the exact minimum number of deletions $\delta(N) = N - f(N)$ by mixed-integer programming (floating-point solver, not certified) and compared it with the right-hand side of (1):

N	exact $\delta(N)$	bound from (1)	bound from (2)	bound from (3)
100	14	11	13	14
150	25	17	20	23
200	33	20	25	30
250	40	26	31	37
300	49	32	39	43

The bounds are valid in every case, as they must be, and the five-prime box recovers most of the true deletion count at these small N.

4 Where this method saturates

Table 1 lists further boxes. Entries marked "LP" use the same procedure as above (smallest threshold at which $\lceil$LP value$\rceil$ increases) computed in floating point; we did not export certificates for them. Entries marked "MILP" use exact integer covering numbers from a floating-point solver and are likewise uncertified.

primes	box	cells	bound on $\limsup f(N)/N$	type
$\{2,3\}$	$(3,2)$	6	0.89286	certified
$\{2,3,5\}$	$(6,5,3)$	90	0.87375	certified
$\{2,3,5\}$	$(6,5,3)$	90	0.86972	MILP
$\{2,3,5\}$	$(8,6,4)$	192	0.87185	LP
$\{2,3,5\}$	$(10,7,4)$	280	0.87149	LP
$\{2,3,5\}$	$(12,8,5)$	480	0.87126	LP
$\{2,3,5,7\}$	$(6,4,3,2)$	144	0.86997	LP
$\{2,3,5,7\}$	$(8,6,4,2)$	384	0.86680	LP
$\{2,3,5,7,11\}$	$(6,4,2,2,2)$	192	0.86727	LP
$\{2,3,5,7,11\}$	$(7,5,3,2,2)$	420	0.86382	certified

Table 1: Bounds from single-level box tilings.

For a fixed prime set the bound improves only slowly as the box grows (for $\{2,3,5\}$ it stalls near 0.871 with LP-rounded covers), while each additional prime helps more, at the price of a much larger box. This is consistent with Kitamura's use of many primes and a recursion on the covering number of the whole smooth set $S_P(x)$ instead of a fixed box. As a further numerical indication (not a theorem and not an asymptotic statement), the LP relaxation of the covering problem on the full interval $[1,3000]$ has value 495.5, so $f(3000) \le 2504$, i.e. $f(N)/N \le 0.835$ at $N = 3000$; this suggests that block decompositions have room below the constants above, in line with the recorded 0.8462.

5 Remarks and limitations

- **No new record.** All three constants in Theorem 4 are weaker than $0.84617\ldots$ [3]. The note is meant as an exposition of the single-level case with checkable certificates, not as an improvement.
- **Lower bound.** Nothing here changes the lower bound $\frac{5}{8}$ of Proposition 1. The true value of $\lim f(N)/N$, if it exists, lies somewhere between $\frac{5}{8}$ and the upper bounds above and is not known to us; the exact values $f(N)/N$ in the table of Section 3 (between about 0.84 and 0.86 for $100 \le N \le 300$) are well above $\frac{5}{8}$, although small-N values say little about the limit.
- **Uncertified entries.** Table 1 entries not marked "certified" have not been checked in exact arithmetic and should be treated as numerical evidence only.
- **Formalisation.** The certificates in Theorem 4(1)–(3) are finite rational data whose verification involves only integer arithmetic; they are suitable for a Lean `decide`-style check. The passage from (1) to the asymptotic statement is a routine counting argument (Rankin's trick and inclusion–exclusion) as given in the proof of Theorem 2.

Reproducibility

The files `box_2_3.json`, `box_2_3_5.json` and `box_2_3_5_7_11.json` contain, for each k, the threshold τ_k and the integer-weighted packing (weights over the scale 10^9). The script `verify.py` (standard library only) regenerates the cells, checks that every listed triple is harmonic, distinct, inside the box and below the threshold, checks all vertex loads and packing totals, and prints D as an exact fraction.

Disclosure and acknowledgments

The computations, the framework presentation and the text of this note were produced by the AI system Claude (Anthropic) in conversation with the submitting author, who supplied the direction and the Lean-verification plan. No independent human mathematician has reviewed the argument. The problem statement, known bounds and attributions are taken from the Erdős Problems website and the cited repository as of October 2026 and should be re-checked before any submission. The author declares the intention to submit this note only after human review.

References

[1] T. F. Bloom, *Erdős Problem #302*, https://www.erdosproblems.com/302.

[2] T. F. Bloom, *Erdős Problem #301*, https://www.erdosproblems.com/301.

[3] K. Kitamura, *A Lean proof of an improved upper bound for Erdős Problem 302*, GitHub repository `KitaKen1/erdos-302-upper-bound` (September 2026); see also pull request #6580 to `google-deepmind/formal-conjectures`.

[4] OEIS Foundation, sequence A390395, https://oeis.org/A390395.